

INTRODUCTION TO SYNTAX I

Session 3: Satisfying syntactic requirements via structure building

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TODAY'S PLAN

- ◉ Combining elements into phrases
- ◉ Recursion
- ◉ Phrase structure
- ◉ Phrase types and terminology

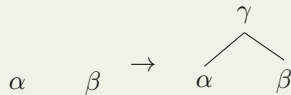
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STRUCTURE BUILDING

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■ Structure-building operation Merge

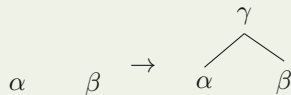
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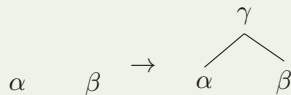
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Syntactic structures must not contain any [uX] features.

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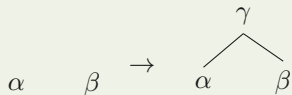
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Syntactic structures must not contain any [uX] features.

- ⦿ How do the [uX] features of predicates 'disappear'?

(2) Feature Checking under sisterhood

A c-selection feature [uX] on a constituent Y is checked (and thus deleted) if Y is the sister of a constituent Z carrying a matching feature [X].

- ⦿ Notation for checked features: $\{\bar{uX}\}$

FEATURE CHECKING: INTRANSITIVE VERBS

(3)

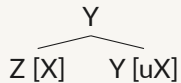
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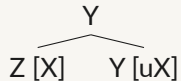


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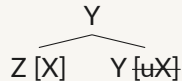
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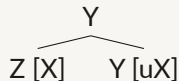


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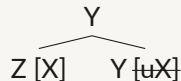
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■ Let's build a simple sentence: Predicate: *arrive* [V, uN]; argument: *the train* [N]

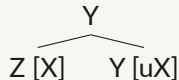
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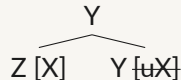
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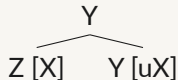
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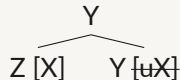
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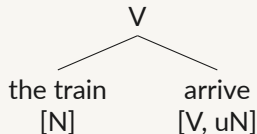
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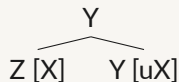


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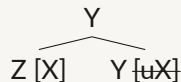
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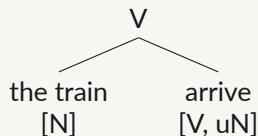
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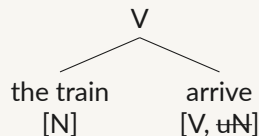
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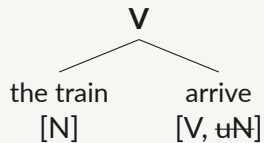


CONSTITUENT LABELS

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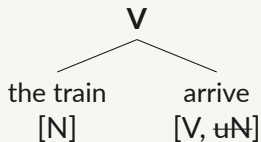
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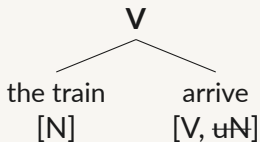
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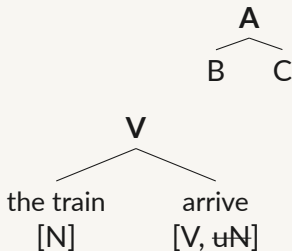
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- ⊙ The label represents the properties of a complex constituent.
- ⊙ The label of a constituent is inherited from one of its **daughter nodes**.
- ⊙ The daughter node from which the label inherited to the mother node is the **head** of the constituent.

■ Inflectional features are inherited from the head

- (12) a. My friend is cheerful.
b. My parents are cheerful.
c. [The **parents** of my friend] are cheerful.
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- a. I ordered desert.
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(15) The verb *order* c-selects a noun:

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■ ⇒ ✓ *will* [V, uV]; ✗ *order* [V, uN] ⇒ *devour* [V] is the head!

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■ The element that c-selects projects!

- ⇒ Heads project their requirements, constraining what syntactic structures can be built.

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- ⊙ Let's see how *Hannah devoured the pizza* is derived:¹

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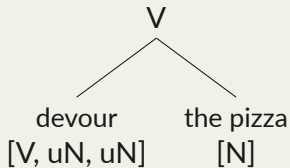
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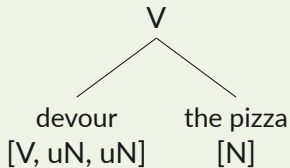
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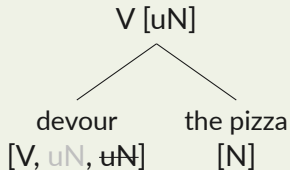
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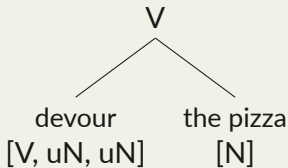
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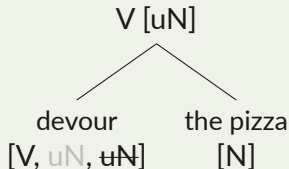
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Step 1: *devour* [V, uN, uN] and *the pizza* [N] are inserted from the numeration.

Step 2: Merge combines *devour* with *the pizza*.

Step 3: one c-selectional feature on *devour* [V, uN, uN] is checked+deleted; the other one is inherited to the mother V.

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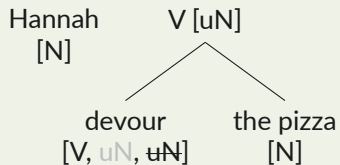
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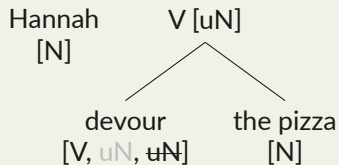
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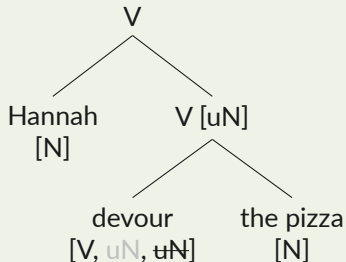
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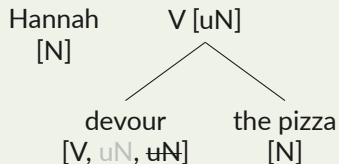
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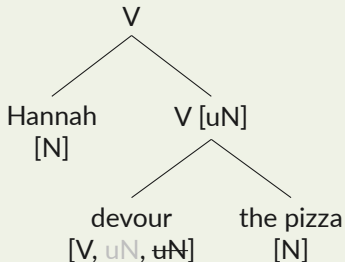
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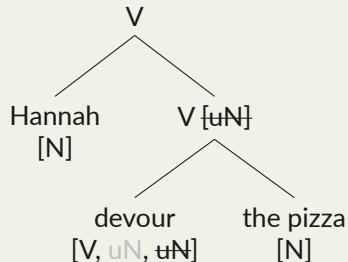
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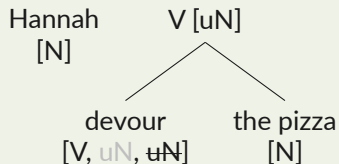
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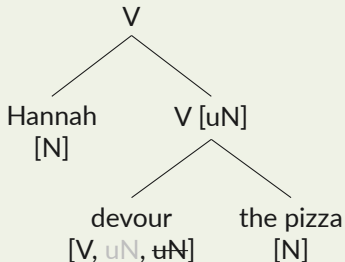
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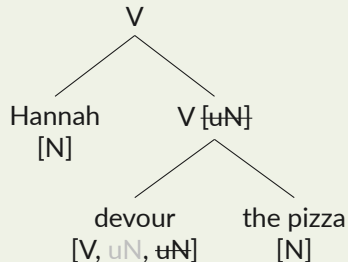
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Step 4: *Hannah* is inserted from the numeration.

Step 5: *Hannah* is merged with *devour the pizza* $[uN]$.

Step 6: the c-selectional feature on *devour the pizza* $[uN]$ is checked+deleted $\Rightarrow$ the derivation is complete.

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■ Formalization via Merge:

- $\Rightarrow$ The complexity of syntactic structures results from repeated applications of Merge.
- $\Rightarrow$ The constituents' selectional criteria constrain what they can Merge with.

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- ⇒ **Valency** independently requires predicates to combine with a specific number of arguments, to ensure that the expressed event is **semantically meaningful**.
 - ⇒ **Selection** is a broader property of heads in general about the **syntactic category of the elements they can combine with**.
 - ⊙ In constituents like *the parents of my friend* (12) and *owners of a pig* (13), the heads *the parents* and *owners* select a preposition, but they could also appear without it.
- (22) a. The parents are responsible for the child.
b. Owners are free to choose the tenants they like.

EXERCISE (10 MIN)

(23) the books on the table

① Together:

- ◇ What is the head?
- ◇ What does the head select?
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② Individually:

- ◇ What is the structure of the whole constituent? Draw a tree introducing each element step-by-step
- ◇ Once you have the tree: label all of the nodes based on category and selectional features.
- ◇ **Hint 1:** no need to overthink the internal structure of *the books* or *the table*.
- ◇ **Hint 2:** if you have trouble figuring out which elements belong 'closer' together, try the constituency tests from session 1!

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- ⊙ it was [***on the table***] that I left the books
- ⊙ *the books* [*on the table*] **and** [*in the library*]

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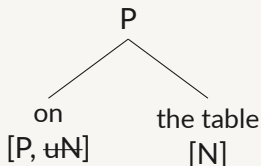
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- *the books* [*on the table*]/[*in the library*]
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- *the books* is the head
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- The whole constituent is a noun

on the table is a constituent!

- *the books* [*on the table*]/[*in the library*]
- *it was* [***on the table***] *that I left the books*
- *the books* [*on the table*] **and** [*in the library*]

(24)

on
[P, uN]

the table
[N]

(25)

P

on
[P, uN]

the table
[N]

(26)

N

the books
[N, uP]

P

on
[P, uN]

the table
[N]

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I Next, we will learn more about 'projection'.

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⇒ In (18), the remaining [uN] projected from *devour* to V.

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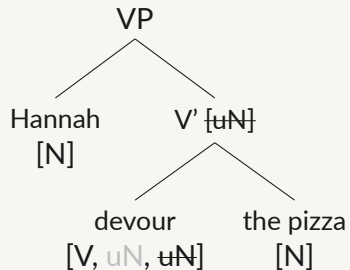
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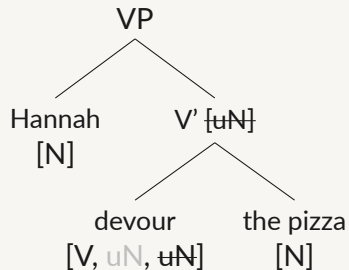
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(27)

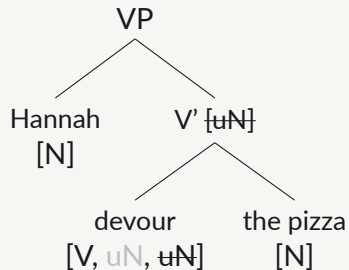


(27)



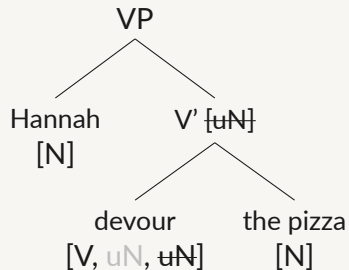
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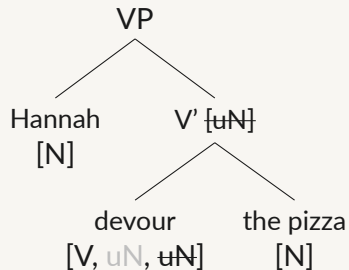
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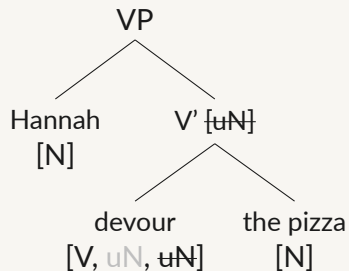
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- ⊙ Implicitly, I showed you that **the smallest possible constituents combine first**:
⇒ In (24-26), *on the table* is built before Merging with *the book*, for example; in (16-18), *devour the pizza* is built before Merging with *Hannah*.

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- ◉ **Merge** combines exactly two elements at a time ⇒ a head has exactly one complement!
- ◉ **Complements are maximal projections** (*the pizza* has no [uX]] features) and **sisters of minimal projections** (*devour* is a minimal projection).

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- ⊙ Ok, so *the pizza* Merges with *devour* first to check one of its [uN] features, becoming its complement. What about *Hannah*?

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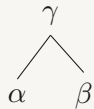
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I A note on linearization

- ◉ The left-to-right order in which our constituents appeared in the hierarchical structures we saw so far happened to correspond to the order in which they linearly appear in the sentence.
- ⇒ This is not always the case!

(28) $[\gamma \ \alpha \ \beta]$

(29)

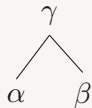


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■ **Merge is insensitive to linear order.**

- ◉ Whether (29)/(28) is linearized as $\alpha \succ \beta$ or $\beta \succ \alpha$ depends on language-specific linearization rules!

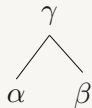
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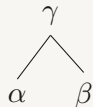
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- ◉ Such rules determine whether a complements and specifiers are linearized to the left or right **relative to their sister node**.

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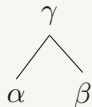


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- ⇒ In other words, a complement is linearized relative to the head (a minimal projection), while a specifier is linearized relative to the [head+complement] (an intermediate projection)!

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- ⇒ In other words, a complement is linearized relative to the head (a minimal projection), while a specifier is linearized relative to the [head+complement] (an intermediate projection)!
- ⇒ **A specifier cannot intrude between a head and its complement!**

Ask away!